

# Electroplating: Copper Coating an Key

Discovering how electricity can transfer a shiny layer of copper metal onto an ordinary iron key!

## 1. Introduction: Why is One Key Dull and the Other Shiny?

Have you ever noticed how a old iron key looks dark, rusty, and dull, while brand-new brass keys or ornaments sparkle with a bright metallic shine? Through the power of electricity, we can take an ordinary dull iron key and coat it with a thin, gleaming layer of copper metal!

## 2. What is Electroplating?

### The Process of Deposit by Electricity

**Electroplating** is the process of depositing a thin layer of a desired metal onto another object using an electric current.

#### Key Components of the Electroplating Setup:

**Anode (+)**

Copper Strip  
(The Copper "Giver")

**Cathode (-)**

Iron Key  
(The Key "Receiver")

**Electrolyte Solution**

Blue Copper Sulphate  
(*"Neela Thotha"* /  $\text{CuSO}_4$ )

### Safety First! "Jugaad" & Lab Safety Tips

- **Protective Gear:** Always wear safety goggles and gloves to protect your eyes and hands.
- **Low Voltage Only:** Use only a small **9V battery**. NEVER connect experiments to wall power sockets!
- **Chemical Care:** Handle Copper Sulphate ( $\text{CuSO}_4$ ) carefully — do not taste, touch directly, or ingest.
- **Clean Workspace:** Keep your experiment area completely clean, tidy, and dry at all times.

## 3. How to Copper Coat a Key: Step-by-Step Procedure

1. **Clean the Iron Key:** Thoroughly scrub the iron key with sandpaper or vinegar to remove rust, oil, and dirt.
2. **Prepare the Electrolyte:** Dissolve copper sulphate crystals in water inside a glass container to form a bright blue *Neela Thotha* solution.
3. **Connect the Anode:** Attach the copper metal strip to the positive terminal (+) wire of the 9V battery.

4. **Connect the Cathode:** Attach the cleaned iron key to the negative terminal (-) wire of the 9V battery.
5. **Submerge in Solution:** Dip both the copper strip and the iron key into the blue copper sulphate solution, ensuring they do not touch each other directly.
6. **Wait and Watch:** Allow electric current to flow for 10–15 minutes. Watch a reddish-brown shiny copper layer form on the key!

## 4. Real-Life Applications of Electroplating

Electroplating is widely used around us every day to enhance appearance and prevent corrosion:

### Temple Ornaments & Puja Items

Gold-plated temple kalash, oil lamps (diyas), and decorative idols.

### Vehicle & Household Fittings

Chrome-plated bicycle handlebars, car bumpers, and shiny bathroom taps.

### Utensils & Hardware

Brass-coated cooking utensils, shiny door handles, and metal locks.

### Artificial Jewellery

Faux gold/silver-coated necklaces, bangles, and copper-plated accessories.

# Practice Questions: Electroplating

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Section A: Multiple Choice Questions

**Q1. Which part of the electroplating setup gives copper to the key?**

- (a) The 9V battery
- (b) The copper strip (Anode)
- (c) The iron key (Cathode)
- (d) The glass beaker

**Q2. To electroplate an iron key with copper, the iron key must be connected to which terminal of the battery?**

- (a) Positive (+) terminal
- (b) Negative (-) terminal
- (c) Neutral terminal
- (d) Either positive or negative terminal

**Q3. What is the common Indian name for blue copper sulphate solution used as the electrolyte?**

- (a) Baking Soda
- (b) Neela Thotha
- (c) Caustic Soda
- (d) Fitkari

**Q4. Why is a 9V battery recommended for this experiment instead of connecting directly to a wall power outlet?**

- (a) 9V battery current is blue in colour
- (b) Wall electricity is low voltage and ineffective
- (c) Wall electricity carries dangerously high voltage and can cause fatal electric shocks
- (d) Batteries produce faster copper melting

## Section B: Short Answer & Thought-Provoking Questions

**Q5. Fill in the blanks with appropriate scientific terms:**

- a) The electrode connected to the positive terminal of the battery is called the \_\_\_\_\_.
- b) The electrode connected to the negative terminal of the battery is called the \_\_\_\_\_.
- c) In this experiment, the electrolyte solution used is \_\_\_\_\_ solution.

**Q6. Why do you think electroplating an iron key with copper is better than simply painting it with copper-coloured paint?**

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**Q7. List two real-life examples where electroplating is used to make objects look attractive or protect them from rusting.**

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**Q8. Suppose a student accidentally connects the iron key to the positive (+) terminal and the copper strip to the negative (-) terminal. What would happen during the experiment?**

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